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2027 CYCLE

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CUET UG

Mathematics

Probability Distributions

Expectation, Variance, Binomial, Poisson and Normal · Section B2 IV · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) 2026 syllabus — Mathematics/Applied Mathematics (319), the current 2027 baseline)	Skills the questions test
Probability distributions of discrete random variables	"Understand the concept of Random Variables and its Probability Distributions. Find probability distribution of discrete random variable." (Section B2 Applied Mathematics Unit IV)	Turning a story into a distribution table; policing the sum-to-one rule.
Mathematical expectation	"Apply arithmetic mean of frequency distribution to find the expected value of a random variable." (Section B2 Applied Mathematics Unit IV)	Computing $E(X) = \Sigma x_i p_i$ and reading it as a long-run average or fair price.
Variance and standard deviation	"Calculate the Variance and S.D. of a random variable." (Section B2 Applied Mathematics Unit IV)	Two-term variance recipes without the μ^2 slip.
Binomial distribution	"Identify the Bernoulli Trials and apply Binomial Distribution. Evaluate Mean, Variance and S.D. of a Binomial Distribution." (Section B2 Applied Mathematics Unit IV)	Verifying the four trial conditions; computing exact and "at least" binomial probabilities.
Poisson distribution	"Understand the conditions of Poisson Distribution. Evaluate the Mean and Variance of Poisson distribution." (Section B2 Applied Mathematics Unit IV)	Recognising rare-event setups; using mean = variance = λ as the identifier.
Normal distribution	"Understand normal distribution is a continuous distribution. Evaluate value of Standard normal variate. Area relationship between Mean and Standard Deviation." (Section B2 Applied Mathematics Unit IV)	Converting $x \to z$, reading 68/95/99.7 areas, using symmetry.

EXAM PATTERN

CUET UG pattern (NTA official information bulletin, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The 2026 syllabus houses this unit as **Section B2 Applied Mathematics Unit IV** — the applied-statistics heart of the paper,

where every formula below is attached to a one-line "when does it fire" test that makes the choice of distribution instant.

Discrete Random Variables — From Story to Table

A random variable X turns experiment outcomes into numbers, and its **probability distribution** is simply the full price list: every value X can take, with its probability. Two rules police every table: each probability lies in $[0, 1]$, and the column sums to **exactly 1**. Those checks come before any computation — several MCQ options are tables that quietly fail them. Building the table from a story is a two-step habit: list the values X can take, then count the outcomes giving each value.

WORKED EXAMPLE — SIXES IN TWO THROWS

A fair die is thrown twice; X = number of sixes. Build the distribution.

Solution. Values: $X = 0, 1, 2$. Each throw: $P(\text{six}) = \frac{1}{6}$, $P(\text{not}) = \frac{5}{6}$.

$$P(X=0) = \frac{5}{6}\cdot\frac{5}{6} = \frac{25}{36}, \quad P(X=1) = 2\cdot\frac{1}{6}\cdot\frac{5}{6} = \frac{10}{36}, \quad P(X=2) = \frac{1}{6}\cdot\frac{1}{6} = \frac{1}{36}.$$

Police check: $25 + 10 + 1 = 36$ — the column sums to 1 ✓. The "$\times 2$" in the middle line is the **two orders** the single six can arrive in; forgetting it is the most common table error in this family.

Mathematical Expectation — the Long-Run Average

The expectation $E(X) = \Sigma x_i p_i$ is a weighted average: each value weighted by how often it shows up. Read it as **the average per trial over a very long run** — a fair price in games, a budgeted cost per item in quality control. The arithmetic-mean bridge in the syllabus is exactly this: for a frequency distribution, the mean is $\Sigma x_i \frac{f_i}{\Sigma} f_i$; with frequencies replaced by probabilities, the same machine computes $E(X)$.

WORKED EXAMPLE — IS THE GAME FAIR?

You pay ₹5 to roll one die. You win ₹12 if it shows six, ₹4 otherwise (your ₹4 returned). Find your expected gain per game.

Solution. Gain X: $+7$ with probability $\frac{1}{6}$ (win 12, paid 5), -1 with probability $\frac{5}{6}$ (get back 4, paid 5). Then

$$E(X) = 7\cdot\frac{1}{6} + (-1)\cdot\frac{5}{6} = \frac{7}{6} - \frac{5}{6} = \frac{2}{6} = \frac{1}{3}.$$

You gain one-third of a rupee per game on average — the game favours you slightly. The trick is always the same: define the **net gain** including the fee, list its distribution, then weight.

Variance and Standard Deviation — the Spread Dial

The mean says where a distribution centres; the **variance** says how widely it scatters around that centre: $\text{Var}(X) = \Sigma x_i^2 p_i - [E(X)]^2$, with the standard deviation its square root. The recipe is strictly two-term — sum the squares weighted, **then** subtract the squared mean. Skipping the subtraction is the most-repeated variance error in every exam season; the inflated answers sit in the options waiting for it.

WORKED EXAMPLE — TWO COINS, FULLY WORKED

X = heads in two fair tosses: distribution $0, 1, 2$ with $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Find variance and SD.

Solution. Mean: $\mu = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 1$. Weighted squares: $\Sigma x_i^2 p_i = 0 + 1 \cdot \frac{1}{2} + 4 \cdot \frac{1}{4} = \frac{3}{2}$. Variance:

$$\sigma^2 = \frac{3}{2} - 1^2 = \frac{1}{2}, \quad \sigma = \frac{1}{\sqrt{2}} \approx 0.71.$$

Three lines, each checkable — and identical machinery works for every discrete table in the chapter.

Bernoulli Trials and the Binomial Distribution

A **Bernoulli trial** is one yes/no experiment: two outcomes, a constant success probability p across trials, and independent trials. Stack n of them and count successes: X follows the **binomial distribution**

$$P(X = r) = \binom{n}{r} p^r q^{n-r}, \quad q = 1 - p, \quad r = 0, 1, ..., n$$

with mean np, variance npq, and standard deviation $\sqrt{npq}$. Before any binomial computation, verify the four conditions — a fixed n, exactly two outcomes, constant p, independence — because "draw **without** replacement from a small bag" breaks the constant-p condition and the formula quietly stops being valid.

- **Exact counts:** "exactly 2 heads in 5 tosses" is $\binom{5}{2}\left(\frac{1}{2}\right)^5 = \frac{5}{16}$ — choose the positions, multiply the probabilities.
- **"At least one" is the complement again:** $P(X \geq 1) = 1 - q^n$. For 5 tosses: $1 - \left(\frac{1}{2}\right)^5 = \frac{31}{32}$ — one line instead of five cases.
- **The mean/variance pair is an identifier:** mean np and variance npq (with $q \neq p$ in general, so they are **unequal**) — never λ, that is the next distribution's signature.

WORKED EXAMPLE — AT LEAST TWO SIXES IN THREE THROWS

A die is thrown 3 times. Find the probability of at least two sixes.

Solution. Binomial with $n = 3$, $p = \frac{1}{6}$, $q = \frac{5}{6}$. "At least two" = exactly two

1. exactly three:

$$P(X=2)=\binom{3}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)=3\cdot\frac{5}{216}=\frac{15}{216},\quad P(X=3)=\left(\frac{1}{6}\right)^3=\frac{1}{216}.$$

Total: $\frac{16}{216}=\frac{2}{27}$. Complement flavour: "at least one six" would be $1-\left(\frac{5}{6}\right)^3=\frac{91}{216}$ — shorter, but it answers a **different** question. Read which one is asked.

Poisson Distribution — Rare Events, One Dial

When n is huge, p tiny, and only the **rate** $\lambda=np$ is quoted (calls per hour, accidents per week, typos per page), the binomial compresses into the **Poisson distribution**:

$$P(X=r)=\frac{e^{-\lambda}\lambda^r}{r!},\quad \text{with mean} = \text{variance} = \lambda.$$

The equal mean-and-variance pair is its fingerprint — the fastest multiple-choice identifier in the chapter. One dial (λ) sets the whole shape: $e^{-\lambda}$ is the probability of a **quiet** window (zero events), and the factorial $r!$ fades fast, so probabilities concentrate near λ.

WORKED EXAMPLE — TWO CALLS AN HOUR

A helpline receives on average 2 calls per minute. Find the probability of exactly 3 calls in a minute, and of no calls.

Solution. Poisson with $\lambda=2$. Exactly 3:

$$P(X=3)=\frac{e^{-2}2^3}{3!}=\frac{8e^{-2}}{6}=\frac{4}{3}e^{-2}\approx 0.180.$$

None: $P(X=0)=e^{-2}\approx 0.135$ — one in seven minutes goes silent. Note how the single rate number λ answered both questions; that economy is why the Poisson model is everywhere in operations.

The Normal Distribution — the Bell With Two Dials

The **normal distribution** is the continuous bell that heights, marks and measurement errors love to follow. Two dials set it: the mean μ (centre) and the standard deviation σ (spread). Three facts do all the exam work. First, it is **continuous**: $P(X=a)=0$ exactly — only **intervals** carry area, so "probability at 170 cm" is always a trick. Second, symmetry: the bell halves around μ. Third, the **area rule**: within $\pm1\sigma$ lies about 68%, within $\pm2\sigma$ about 95%, within $\pm3\sigma$ about 99.7% of the data.

The **standard normal variate** converts any bell to the universal scale:

$$z=\frac{x-\mu}{\sigma}$$

— a pure number counting "how many standard deviations from the mean". Once converted, standard tables (and the area rule) answer everything; a below-average x produces a negative z, which is information, not an error.

WORKED EXAMPLE — MARKS TO Z, AND BACK

Marks follow a normal distribution with $\mu = 60$, $\sigma = 10$. Find the z-score of 70, the z-score of 45, and the approximate share of students scoring between 50 and 70.

Solution. $z = \frac{70-60}{10} = 1$ — one standard deviation above. $z = \frac{45-60}{10} = -1.5$ — one and a half below, negative because 45 sits under the mean. The band $50 \to 70$ is $\mu \pm 1\sigma$: the area rule says about **68%** of students. Two conversions and one memorised band — the entire normal-distribution question family in miniature.

EVERYDAY EXAMPLE — THREE KINDS OF QUEUE

Binomial is a **bowling** question: a fixed number of throws, each strike-or-miss with the same chance, count the strikes. Poisson is a **rain-leak** question: you cannot count the "trials" of every water molecule, but drips arrive at a steady rate — count drips per hour, λ. Normal is a **crowd height** question: individuals differ, but the crowd stacks into a bell around the average with a predictable spread. Identify which queue you are standing in, and the correct formula walks up to meet you.

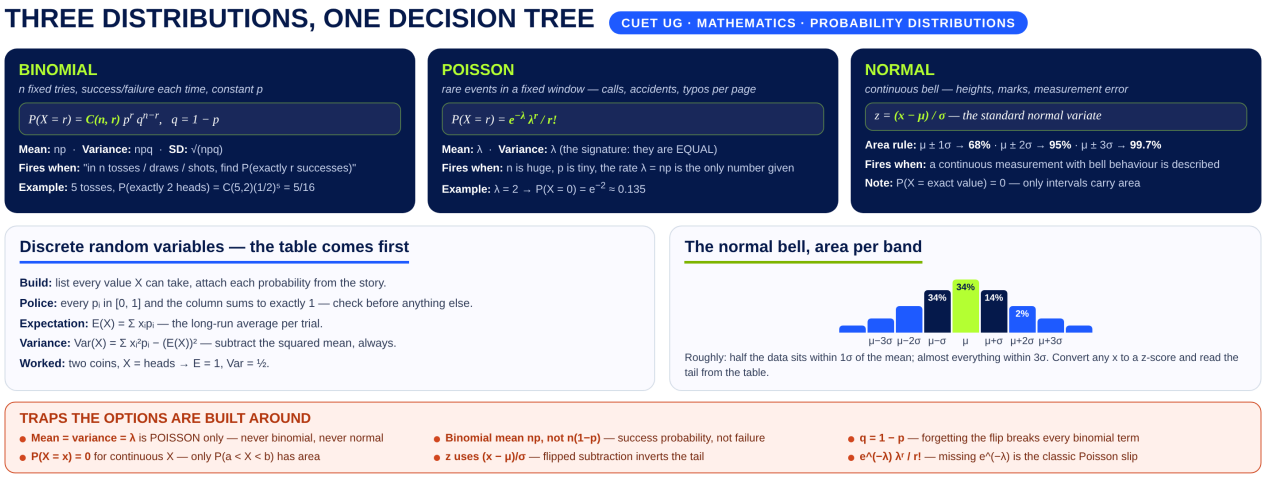


Figure 1: The three distributions side by side: trigger conditions, formulas, mean/variance signatures — and the traps each one plants.

Formula Cheatsheet — the Whole Chapter on One Page

FORMULA CHEATSHEET — PROBABILITY DISTRIBUTIONS

Formula / rule	Letters mean	Fire it when
Table police: $0 \le p_i \le 1$, column sums to 1	before any computing	every distribution table
$E(X) = \Sigma x_i p_i$	weighted average	expectation / fair price
$\text{Var}(X) = \Sigma x_i^2 p_i - \mu^2$	two terms, always both	spread questions
Binomial check: fixed n, two outcomes, constant p, independent	four conditions	before binomial use
$P(X = r) = \binom{n}{r} p^r q^{n-r}$	$q = 1 - p$	exactly-r successes
Binomial mean np, variance npq	SD $= \sqrt{npq}$	binomial summaries
"At least one" $= 1 - q^n$	complement shortcut	repeated trials
Poisson: $P(X = r) = \frac{e^{-\lambda}\lambda^r}{r!}$	rare events, rate λ	huge n, tiny p
Poisson mean = "variance" $= \lambda$	the fingerprint	identifying the model
Normal: two dials μ, σ; bell symmetric	continuous	measurement data
Area rule: $\pm1\sigma \to 68\%$, $\pm2\sigma \to 95\%$, $\pm3\sigma \to 99.7\%$	bell bands	interval shares
$z = \frac{x-\mu}{\sigma}$	how many SDs from centre	standardising any x
$P(X = a) = 0$ for continuous X	only intervals carry area	point-probability tricks

Common Errors and Special Cases

COMMON ERRORS AND SPECIAL CASES — DISTRIBUTIONS

The mistake	What is right	The trap it sets
Using a table that does not sum to 1	Police the column first	Broken tables as "given data"
Variance without the μ^2 subtraction	$\Sigma x_i^2 p_i - \mu^2$, both terms	Inflated numeric options
Binomial applied without replacement	Constant p needs with-replacement or large populations	Small-bag draws dressed as binomial

The mistake	What is right	The trap it sets
Binomial mean as $n(1-p)$	np uses the success probability	Swapped mean options
"Mean = variance" claimed for binomial	Only Poisson carries that signature	The fastest identifier, misassigned
Dropping $e^{-\lambda}$ in Poisson	Every term keeps it	Lambda-only distractor
Point probabilities on a normal bell	$P(X=a)=0$; intervals only	"P(X = 170) = ?" trick stems
z-score with flipped subtraction	$z=\frac{x-\mu}{\sigma}$, sign is information	Negative-z declared "impossible"

Sixty-Second Self-Test

Statement	Verdict
A distribution table may sum to 0.95.	False — exactly 1, always
The binomial mean of $n=10$, $p=0.3$ is 3.	True — np
For Poisson, mean and variance both equal λ.	True — the fingerprint
$P(X=2)$ for a normal X is a small positive number.	False — exactly 0, continuous
$x=65$ with $\mu=50$, $\sigma=5$ gives $z=3$.	True — three SDs above
Without replacement from a small bag is binomial.	False — p changes each draw

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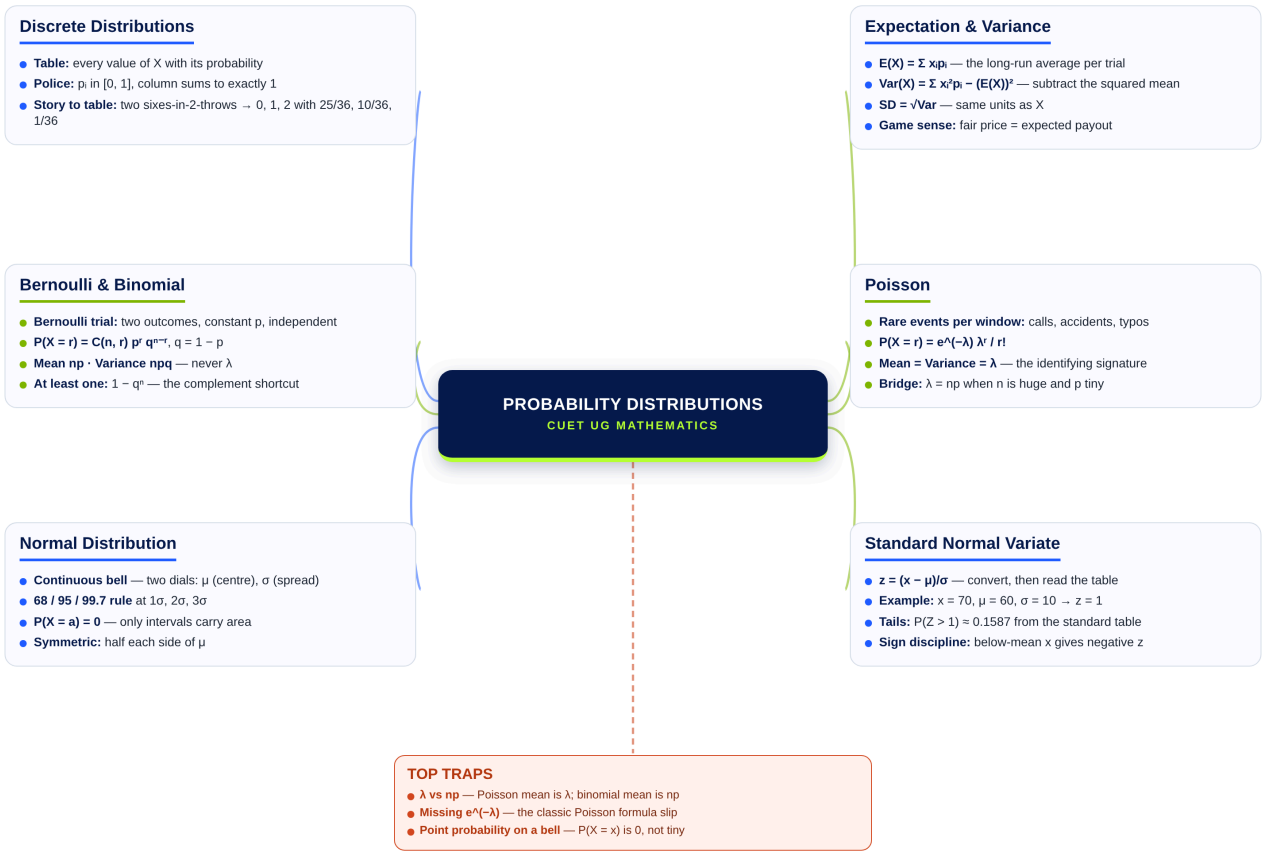


Figure 2: The chapter in one look: tables, expectation, variance, three distributions — and the traps each carries.

Practice Questions — CUET UG Pattern (10 MCQs)

Ten questions in the CUET UG style. Each correct answer earns +5; each incorrect one costs −1. Name the distribution first — the formula follows the name.

Q1. A fair coin is tossed 5 times. The probability of exactly 2 heads is:

(A) $\frac{3}{8}$ (B) $\frac{1}{2}$
(C) $\frac{5}{16}$ (D) $\frac{5}{8}$

Q2. For a binomial distribution with $n = 10$ and $p = 0.3$, the mean and variance are respectively:

(A) 3 and 2.1 (B) 3 and 3
(C) 2.1 and 3 (D) 0.3 and 0.21

Q3. The distribution for which the mean and variance are equal is:

(A) binomial (B) normal
(C) any symmetric distribution (D) Poisson

Q4. Calls arrive at a rate of 2 per minute (Poisson). The probability of no calls in a minute is:

(A) $2e^{-2}$ **(B)** e^{-2}
(C) e^{2} **(D)** $\frac{1}{2}$

Q5. A fair coin is tossed 5 times. The probability of at least one head is:

(A) $\frac{1}{2}$ **(B)** $\frac{5}{32}$
(C) $\frac{31}{32}$ **(D)** $\frac{4}{5}$

Q6. Marks are normal with $\mu = 50$, $\sigma = 5$. The z-score of a student scoring 65 is:

(A) 1.5 **(B)** 3
(C) 5 **(D)** 15

Q7. For a normal distribution, the approximate share of data within $\mu \pm 2\sigma$ is:

(A) 95% **(B)** 68%
(C) 99.7% **(D)** 50%

Q8. A die is thrown twice and X = number of sixes. Then $P(X = 1)$ equals:

(A) $\frac{1}{36}$ **(B)** $\frac{25}{36}$
(C) $\frac{1}{6}$ **(D)** $\frac{5}{18}$

Q9. A random variable has $E(X) = 2$ and $\Sigma x_i^2 p_i = 6$. Its variance is:

(A) 8 **(B)** 2
(C) 4 **(D)** 6

Q10. A fair die is rolled once and X is the number shown. $E(X)$ equals:

(A) 3 **(B)** 3.25
(C) 3.5 **(D)** 4

Detailed Solutions

Sol. Q1. Binomial: $P(X = 2) = \binom{5}{2}\left(\frac{1}{2}\right)^2\left(\frac{1}{2}\right)^3 = 10 \cdot \frac{1}{32} = \frac{10}{32} = \frac{5}{16}$. Choose the two head-positions ($\binom{5}{2} = 10$), then the probability of any fixed pattern is $\frac{1}{32}$.

Answer: (C)

Sol. Q2. Binomial summary pair: mean $= np = 10 \cdot 0.3 = 3$; variance $= npq = 10 \cdot 0.3 \cdot 0.7 = 2.1$. The unequal pair is itself the identifier — equal mean and variance would mean Poisson.

Answer: (A)

Sol. Q3. The mean-equals-variance signature belongs to the Poisson distribution alone. Binomial variance carries the extra q; the normal has two independent dials.

Answer: (D)

Sol. Q4. Poisson with $\lambda = 2$: $P(X = 0) = \frac{e^{-2}2^0}{0!} = e^{-2}$ — the "quiet window" probability. Options with e^2 or $2e^{-2}$ broke the formula's $e^{-\lambda}$ or the exponent of λ.

Answer: (B)

Sol. Q5. Complement route: $P(\text{at least one}) = 1 - P(\text{none}) = 1 - \left(\frac{1}{2}\right)^5 = \frac{31}{32}$. Option (B) is $P(\text{exactly one})$ — the usual neighbour in the option list.

Answer: (C)

Sol. Q6. Convert: $z = \frac{65-50}{5} = \frac{15}{5} = 3$ — three standard deviations above the mean, deep into the tail. The raw gap 15 is not the z-score; dividing by σ is the whole step.

Answer: (B)

Sol. Q7. The area rule: $\pm 1\sigma \to 68\%$, $\pm 2\sigma \to 95\%$, $\pm 3\sigma \to 99.7\%$. Two bands out — 95%. These three numbers answer a dozen question shapes; keep them together.

Answer: (A)

Sol. Q8. Distribution: $P(X = 0) = \frac{25}{36}$, $P(X = 1) = 2 \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{10}{36} = \frac{5}{18}$, $P(X = 2) = \frac{1}{36}$. The middle line carries the $\times 2$ for the two orders the six can arrive in.

Answer: (D)

Sol. Q9. Two-term recipe: $\text{Var}(X) = \Sigma x_i^2 p_i - (E(X))^2 = 6 - 4 = 2$. Option (D) reported the weighted-square sum alone — the classic missing subtraction.

Answer: (B)

Sol. Q10. Expectation of a fair die: $\frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$. Symmetry puts the mean at the midpoint of the values — the sanity check that catches every arithmetic slip.

Answer: (C)

Wrong-Answer Clinic

If you picked this	The thinking slip	The lasting fix
(A) in Q2	Assumed mean = variance for every model	Equal pair = Poisson only; binomial carries the q
(C) in Q4	Kept λ but dropped $e^{-\lambda}$	Write the full Poisson term before substituting

If you picked this	The thinking slip	The lasting fix
(B) in Q5	Computed "exactly one" instead	Underline "at least" — it triggers the complement reflex
(C) in Q6	Reported the raw gap as the score	z divides by σ — it counts deviations, not marks
(D) in Q9	Skipped the μ^2 subtraction	Variance is a two-term recipe; say both terms aloud
(A) in Q10	Averaged badly or forgot the midpoint check	Symmetric values: mean sits at the centre — 3.5 for a die

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